



Moments and Equilibrium of Rigid Bodies

· Mark Scheme

Mechanics Chapter 5: Moments and Equilibrium of Rigid Bodies. M = method mark A = accuracy mark (depends on M)
B = independent mark ft = follow through

Section A – Moments of a force

- 1 12×0.5 M1; = 6 N m A1.
- 2 $5 \times 1 + 8 \times 3$ M1; = 29 N m clockwise A1.
- 3 Perpendicular distance $2 \sin 30^\circ$ M1; $20 \times 2 \sin 30^\circ$ M1; = 20 N m A1.
- 4 Clockwise: $15 \times 2 + 12 \times 4 \sin 30^\circ = 30 + 24$ M1; anticlockwise: $8 \times 6 = 48$ M1; resultant 6 N m clockwise A1.
- 5(a) $20 \times 1 + 3F = 30 \times 2$ M1 A1; $F = \frac{40}{3} = 13.3$ A1.
- 5(b) $30 - 20 - 13.3$ M1; 3.33 N downwards A1.
- 5(c) For equilibrium the resultant force must also be zero B1.

Section B – Equilibrium of a rigid body

- 6 By symmetry, $R = \frac{4 \times 9.8}{2} = 19.6$ N at each end M1 A1.
- 7 $M(A): 3R_C = 196 \times 2$ M1; $R_C = 131$ N A1; $R_A = 196 - 130.7 = 65.3$ N A1.
- 8 $30g \times 1.2 = 24g \times d$ M1 A1; $d = 1.5$ m on the other side A1.
- 9 $R_B = 98 - 40 = 58$ N B1; $M(A): 58 \times 5 = 98x$ M1 A1; $x = 2.96$ m A1.
- 10(a) $R = 100g = 980$ N B1.
- 10(b) $M(A): S \times 6 \sin 60^\circ$ M1; $= 20g \times 3 \cos 60^\circ + 80g \times 4.5 \cos 60^\circ$ M1 A1; $S = 396$ N A1.
- 10(c) $F = S = 396$ N B1; $F \leq \mu R$ M1; $\mu \geq \frac{396}{980} = 0.404$ A1.
- 10(d) The man is a particle (his weight acts at a single point) B1.

Section C – Sliding and toppling

- 11 $F_{\max} = 0.5 \times 100 = 50$ N M1; $40 < 50$, so the box does not move A1.
- 12 About to topple: reaction acts at the front edge M1; $M(\text{edge}): 1.2P = 150 \times 0.3$ M1; $P = 37.5$ N A1.
- 13 Slides when $P = 0.3 \times 150 = 45$ N M1 A1; $37.5 < 45$, so it topples first A1.
- 14 About to tilt about D: $R_C = 0$ M1; $M(D): 10g \times 1 = 15g \times x$ M1 A1; $x = \frac{2}{3} = 0.667$ m A1.
- 15(a) $R_C + R_D = 105g = 1029$ B1; $M(C): 3R_D = 1029 \times 2$ M1; $R_D = 686$ N, $R_C = 343$ N A1.
- 15(b) About to tilt about D, $R_C = 0$ M1; $M(D): 75g \times y = 30g \times 1$ M1; $y = 0.4$ m past D, so 1.6 m from B A1.
- 15(c) $M(D)$ with man at B: $75g \times 2 = 30g \times 1 + mg \times 4$ M1; $m = 30$ kg A1.

Roar-level challenge

- 16 $R = W$, $S = F = \mu R = \mu W$ M1 A1; moments about the foot (length $2a$): $S \times 2a \sin \theta = W \times a \cos \theta$ M1; $2\mu \tan \theta = 1 \Rightarrow \tan \theta = \frac{1}{2\mu}$ A1; $\tan \theta = 1.25$, $\theta = 51.3^\circ$ A1.